

# Java Spring Boot 4 Backend – UML Class Diagram

Spring Boot 4.1.1 • Spring AI 2.0.1 Dual Models (Llama 3.3 & Mistral) • MongoDB Repository • Qdrant Vector Retrieval

PACKAGE: com.personallibrary.controller

**<<RestController>>  
DocumentController**

- documentService: DocumentService  
- extractionService: BibTeXExtractionService  
- summarizationService: AiSummarizationService  
- vectorRagService: VectorRagService  
- storageService: StorageService

+ getDocuments(fileName, title, author, edition, format, content, page, pageSize, sortBy, sortOrder): ResponseEntity<PaginatedResponse<DocumentResponse>>  
+ getDocument(guid: String): ResponseEntity<DocumentResponse>  
+ uploadDocument(file: MultipartFile, metadataJson: String): ResponseEntity<DocumentResponse>  
+ overwriteDocumentVersion(guid: String, file: MultipartFile, metadata: String): ResponseEntity<DocumentResponse>  
+ extractMetadata(payload: Map<String, String>): ResponseEntity<BibTeXMetadata>  
+ deleteDocument(guid: String): ResponseEntity<Map<String, Object>>  
+ regenerateSummary(guid: String, request: SummarizeRequest): ResponseEntity<SummaryRecord>  
+ chatWithDocument(guid: String, request: ChatRequest): ResponseEntity<ChatResponse>

calls

PACKAGE: com.personallibrary.service

**<<Service>>  
DocumentService**

- documentRepository: DocumentRepository  
- storageService: StorageService  
- extractionService: BibTeXExtractionService  
- summarizationService: AiSummarizationService  
- vectorRagService: VectorRagService

+ uploadDocument(file: MultipartFile, userBibtex: BibTeXMetadata): DocumentResponse  
+ overwriteDocument(existingGuid: String, newFile: MultipartFile, updatedBibtex: BibTeXMetadata): DocumentResponse  
+ searchDocuments(criteria: DocumentQueryCriteria, pageable: Pageable): PaginatedResponse<DocumentResponse>  
+ getDocumentByGuid(guid: String): DocumentResponse

**AiSummarizationService**

- llamaChatClient: ChatClient  
- mistralChatClient: ChatClient

+ generateDualSummaries(title, text, bibtex): Map  
+ generateSummaryForModel(modelKey, title, text, bibtex): SummaryRecord

**VectorRagService**

- vectorStore: VectorStore  
- llamaChatClient: ChatClient

+ indexDocumentChunks(docGuid, text): List<Chunk>  
+ chatWithDocument(docEntity, request): ChatResponse

**BibTeXExtractionService**

+ extractMetadata(fileName, sampleContent): BibTeXMetadata

**StorageService**

+ storeFile(file: MultipartFile, guid: String): Path  
+ extractTextContent(path: Path): String  
+ deletePhysicalAsset(guid): void

PACKAGE: com.personallibrary.repository

**<<interface>>  
DocumentRepository**

+ findByGuid(guid: String): Optional<DocumentEntity>  
+ findByPreviousVersionGuid(guid): List<DocumentEntity>

**<<CustomMongoRepository>>  
DocumentRepositoryCustomImpl**

- mongoTemplate: MongoTemplate

+ searchDocuments(fileName, title, author, edition, format, content, pageable): Page<DocumentEntity>

persists / queries

PACKAGE: com.personallibrary.model

**<<Document (collection="documents")>>  
DocumentEntity**

- id: String  
- guid: String  
- previousVersionGuid: String  
- versionNumber: int  
- fileName: String  
- fileSize: long  
- fileSizeFormatted: String  
- format: String  
- uploadDate: Instant  
- editDate: Instant  
- bibtex: BibTeXMetadata  
- bibtexRaw: String  
- summaries: Map<String, SummaryRecord>  
- contentExcerpt: String  
- fullContent: String  
- chunks: List<DocumentChunk>  
- versionHistory: List<DocumentVersionSnapshot>

**<<ValueObject>>  
BibTeXMetadata**

- entryType: BibTeXType  
- bibKey: String  
- title: String  
- author: String  
- year: String  
- journal: String  
- booktitle: String  
- volume: String  
- publisher: String  
- doi: String  
- keywords: String

+ toRawBibTeX(): String  
+ validateMandatoryFields(): boolean

**SummaryRecord**

- modelName: String  
- modelKey: String  
- summaryText: String  
- createdAt: Instant  
- durationSeconds: double  
- durationFormatted: String

**DocumentChunk**

- id: String  
- chunkIndex: int  
- text: String  
- embeddingVector: float[]

**<<enum>> BibTeXType**

ARTICLE, BOOK, INPROCEEDINGS  
TECHREPORT, PHDTHESIS, MISC